Literature Status for Critical Directed Distances in arXiv:1606.04687

Lane 4 autonomous mathematical discovery run

August 11, 2026

Source questions

Brezis–Mironescu–Shafrir, arXiv:1606.04687, define

$$\mathrm{Dist}_{W^{s,p}}(\mathcal{E}_{d_1}, \mathcal{E}_{d_2}) := \sup_{f \in \mathcal{E}_{d_1}} \inf_{g \in \mathcal{E}_{d_2}} |f - g|_{W^{s,p}}.$$

Open Problem 1 on page 5 asks whether this directed distance is symmetric, or even depends only on $|d_1 - d_2|$. Open Problem 2 on the same page asks whether, in the critical space $W^{N/p,p}(\mathbb{S}^N; \mathbb{S}^N)$,

$$\mathrm{Dist}_{W^{N/p,p}}(\mathcal{E}_{d_1}, \mathcal{E}_{d_2}) \geq C'_{p,N}|d_1 - d_2|^{1/p}.$$

The source singles out $(N, p) = (1, 2)$ and $(2, 2)$ as cases of special interest. Its Remark 7.7 records that the directed lower bound was still unknown for $N = 1$, $p = 2$, and $0 < d_2 < d_1$.

Complete answer in the circle-critical family

Shafrir, arXiv:1707.05764, Theorem 1.1, proves for every $1 < p < \infty$ and all $d_1, d_2 \in \mathbb{Z}$ that

$$\mathrm{Dist}_{W^{1/p,p}}(\mathcal{E}_{d_1}, \mathcal{E}_{d_2}) = \sigma_p(d_2 - d_1), \qquad \sigma_p(d) := \inf_{h \in \mathcal{E}_d} |h|_{W^{1/p,p}}.$$

Complex conjugation shows $\sigma_p(-d) = \sigma_p(d)$, so the formula is symmetric and depends only on $|d_1 - d_2|$. The degree-energy estimate and the bubble upper bound give constants $c_1(p), c_2(p) > 0$ such that

$$c_1(p)|d_1 - d_2|^{1/p} \leq \mathrm{Dist}_{W^{1/p,p}}(\mathcal{E}_{d_1}, \mathcal{E}_{d_2}) \leq c_2(p)|d_1 - d_2|^{1/p}.$$

Thus the $N = 1$ instances of both source problems are answered affirmatively. In particular, Corollary 1.1 gives the exact formula

$$\mathrm{Dist}_{H^{1/2}}(\mathcal{E}_{d_1}, \mathcal{E}_{d_2}) = 2\pi|d_1 - d_2|^{1/2}$$

for all integer degrees, including the source's missing direction $0 < d_2 < d_1$.

Additional first-order critical answer

Frank–Ivanisvili, arXiv:2606.29382, Theorem 1.1, proves the sharp formula for the directed distance in $W^{1,N}(\mathbb{S}^N; \mathbb{S}^N)$ for every N. In their normalization both directions equal $(\kappa_N|d_1 - d_2|)^{1/N}$. This resolves the source's highlighted $(N, p) = (2, 2)$ case and establishes symmetry throughout the first-order critical family.

Scope

This is a scoped literature-status identification, not a new proof. The supporting papers do not settle every fractional or higher-order $W^{N/p,p}$ instance in dimensions $N \geq 2$, nor the separate $W^{2,1}(\mathbb{S}^1; \mathbb{S}^1)$ question in the source.

References

[1] H. Brezis, P. Mironescu, and I. Shafrir, *Distances between homotopy classes of* $W^{s,p}(\mathbb{S}^N; \mathbb{S}^N)$, arXiv:1606.04687.

[2] I. Shafrir, *On the distance between homotopy classes in* $W^{1/p,p}(\mathbb{S}^1; \mathbb{S}^1)$, arXiv:1707.05764, Theorem 1.1 and Corollary 1.1.

[3] R. L. Frank and P. Ivanisvili, *The distance between homotopy classes of Sobolev maps on spheres*, arXiv:2606.29382, Theorem 1.1.